# Manual

## PCC Module File Interface Machine/Process Data

## PCC-DIF 8.1

Version 1.0.23049

Last changed on: 02.09.2020

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# Contents

# 1 PCC-DIF Machine Connection

## Summary

The universal file interface **PCC-DIF** of HYDRA PCC (Process Communication Controller) is one component for the connection of machines. This component helps customers or machine manufacturers to easily implement a machine interface on the basis of file transfer. This document describes the configuration of the interface and the file handling including the file structure.

## Fields of application

The file interface PCC-DIF currently supports the following data types and modules:

- MDE data: Counter channels including cycle time, digital I/O and machine status MSTAT

- PDV data: process value channels and trigger channels

In addition, the PCC-ADP (PCC Application Adapter) expansion kit allows for the modules HYDRA-ADE and HYDRA-MPL to be connected. In this context, data and communication processes need to be configured, which is to be defined in a customizing concept and ADP configuration.

## Implementation notes

Communication is based on file transfer. Consequently, the interface can be used everywhere where the machine control can easily access a shared directory by network share or FTP communication and where the control is a system that can easily handle file operations, e.g. servers based on PC.

## Integration

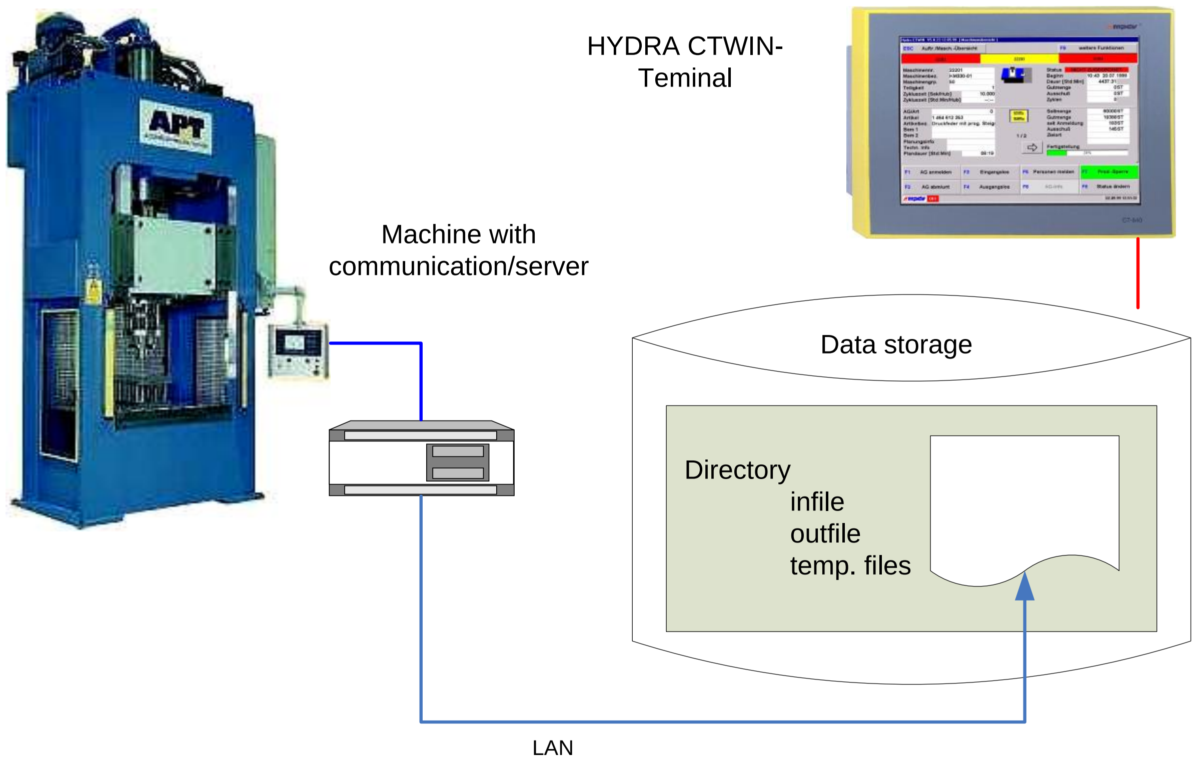
The protocol module is integrated in HYDRA PCC. Please also see the manuals dealing with the products SCS-PCP and SCS-PCB

## Features

- File-based interface to transfer process data and machine data provided by machine controls (PLC), data concentrators, external systems or similar; required once for each machine or aggregate.

# 2 Protocol Module PCC-DIF

## 2.1 Architecture, Data Contents




The file interface is connected with a HYDRA CTWIN terminal. An interface directory used for the communication is defined in the data storage (hard disk) of the CTWIN terminal or in another directory within a network which can be accessed by the terminal. The communication partners' data is stored into - or read out from this directory.

## 2.2 File Handling

Data has to be transferred in a secured process: All read or write operations have to be executed as separate, single processes. For this reason, the copy command provided by the operating system is not appropriate. The rename/move commands have to be used instead, i.e. HYDRA and the partner program each work on temporary copies and separately provide or collect the data using the "rename" function. In case of file conflicts, which might rarely occur though, the error needs to be rectified and the access attempt is to be repeated after an appropriate wait time has elapsed.

HYDRA-PCC-DIF works as follows:

**Receive data:** the existence of the defined data file is constantly checked. If this file is available, it is automatically renamed as a temporary file. HYDRA then internally processes this file entirely. Afterwards the file is deleted. In case the file is still available after restarting the system, it will be processed at first.

**Send data:** Data is attached (appended) inside the existing, temporary outbound file or a new file is created. Once the write process has been finished, this file is renamed as the final outbound file, provided that this outbound file does not exist at that moment. If the outbound file exits, it is waited for the partner to pick up this file. During this wait time, data is continued being attached (appended) within the existing temporary file.

**The partner system has to comply with this procedure. This ensures that no conflicts occur and no data gets lost.**

**Communication monitoring – error handling:**

The following PCC-DIF procedure may be used to monitor communication: A timeout can be defined for outbound as well as inbound files, which specifies when an outbound file has to be collected or a new inbound file has to be available. If the timeout has been exceeded, the PCC-DIF driver interprets this as an error in the communication and returns it to the terminal/shop floor PC. Consequently, the interface function can be represented as channel fault at the terminal. Operating system errors occurred during the read/write process are also sent as error in the communication to the terminal. If timeouts are defined, at least empty files have to be exchanged between the two communication partners. Vice versa, it might also be important to the communication partner to recognize whether or not HYDRA is ready to communicate. This is ensured by HYDRA providing at least an empty file, which can be picked up by the partner, within a defined interval. The driver creates this file as substitute, unless a communication takes place anyway during this interval.

## 2.3 File Structure

The files are text files including at least one data record per line. This allows for several data records to be transferred in a sequential order within one file. The file is processed according to the FIFO principle. The file is read out starting at the beginning; new data records are attached at the end.

Each line has the following dynamic structure:

<Identifier>=<Value>|…

Example:

```
COUNTER1=25655|COUNTER2=7625|INP01=1|ZYK1=12.5|P1=2345.31|P2=231.1|DAT=08/30/2006|ZEI=16:54:20
COUNTER1=25656|COUNTER2=7625|INP01=1|ZYK1=12.3|P1=2456.34|P2=231.8|DAT=08/30/2006|ZEI=16:54:32
COUNTER1=25657|COUNTER2=7627|INP01=0|ZYK1=12.2|P1=2645.65|P2=231.6|DAT=08/30/2006|ZEI=16:54:44
…
```

Such lines are listed one after the other; each line may have different contents. The respective configuration determines whether or not this delivers reasonable results.

If several groups are defined at collection channels (see below), different files have to be used.

## 2.4 File Content

The driver imports the data file lines one after the other and saves them in internal line storage on the group channels.

This data is transferred to the HYDRA applications on the basis of the specified read-out rules, i.e. data included in one line is interpreted together:

### 2.4.1 Machine Data

Operating statuses of machines and quantity information rank among the shop floor data provided by the HYDRA-MDE area. Digital information on malfunctions as well as meter readings of counter registers may be transferred. Downtime reasons may also be forwarded as value (MSTAT) in a table.

Examples:

……..|BETRIEBSIGNAL=0|E_STOERUNG=1|………….. → Machine down, electrical interference.

……..| BETRIEBSIGNAL=1|E_STOERUNG=0|………….. → Machine is producing

……..|ZAEHLER_IO=1234|………….. → current meter reading
……..|ZAEHLER_IO=1235|………….. → meter reading has been incremented by 1, the number of pieces
in HYDRA will also be increased by 1.

…|Maschinenstatus|…
……..|MSTAT@M100=1| → the machine is in machine status: 1 (this could be production)
…...|MSTAT@M100=25| → new machine status: 25 (this could be a malfunction according
to the configuration)

Contents of counter registers are transferred as counter information. HYDRA only evaluates counter differences as increase in the quantities.

The below definition applies:

- The counter register increases up to a meter reading **OVERFLOW** that is defined in the driver's INI file and restarts at an initial value (e.g. OVERFLOW = 30000). Only a global value may be defined. All counter registers within a file work with the same OVERFLOW value. If a lower figure is suddenly imported, it is grossed up to the overflow and counted from the initial value

- The counter register starts at a meter reading **OVERFLOW_MODE** defined in the driver's INI file**.** Three statuses can be distinguished:
  1. OVERFLOW_MODE=0 after an overflow it is restarted at '0'.
  2. OVERFLOW_MODE=1 after an overflow it is restarted at '1'.
  3. OVERFLOW_MODE=RESET if a lower value is recognized
  the last imported figure is considered being the overflow and
  the meter readings until this overflow has been reached are ignored.

Once the terminal application "ctwin" has been restarted, the counter value of a counter register that is first identified is used as new reference value for determining the differences. Only when this counter register has been identified once more, the counter difference can be forwarded to HYDRA.

### 2.4.2 Transfer of Values

If values, such as weights or article numbers (no process data) are to be transferred from a file to HYDRA, they may be included as values in data records. The values are transferred in configuration "V": the separator "|" must not appear in the transfer values.

### 2.4.3 Process Data (for the HYDRA-PDV module)

Process data is read out from the line as tuple of measured values and saved in an internal buffer belonging to the module. Depending on how the read-out is configured (by trigger or cyclically), HYDRA accepts the saved process values. When data is to be collected by trigger, the trigger should be provided in the same line where the PDV values are defined. As in any other case, the PDV values of the previous line that has been imported will be transferred to HYDRA. If cyclic data collection is used, the PDV values, which are stored within the buffer at the time of the request, will only be transferred to HYDRA.

A buffered transfer of the rows may be used for transferring mass data (high volume data). In the configuration file of the driver, the item "PDV_MODE=AUTO" is set ("automatic"). Now any process data is included in the automatic signal processing of HYDRA. If "PDV_MODE=MAN" is set (manual), the above-described procedure will be reset and HYDRA will control the data transfer manually.

**Time monitoring:**

The data lines may be provided with a time stamp by indicating DATE (notation yyyy/mm/dd) and TIME (hh:mm:ss). As far as it is possible, this time stamp is taken into account for postings. In particular in the automatic mode, each line including its time stamp is transferred to HYDRA-PDV data collection. However, please note in this context that the HYDRA clock never goes backwards, i.e. the times must be ascending. Moreover, current events may affect the posting of old values in a way which prevents them from being taken into account, e.g. collision of manual postings of the machine status with still pending automatic postings. Furthermore, cycle time monitoring can only be used online! Process data may have an old time stamp but online warnings are no longer possible.

Thus, it is reasonable to keep the connection up-to-date to avoid errors in data collection and posting due to time delays.

Measured values are reworked statistically in HYDRA-PDV 7.2. But this is only possible for a limited period of time (reaching into the past) when it comes to data collection. Older measured values are no longer recalculated statistically. By default, this is one hour, i.e. the time stamps included in the data rows must not be older than one hour. The data transmitter, machine or interface, has to guarantee this.

If no time stamp is defined, the system always refers to the current time of the data import.

## 2.5 Channel Concept

The transfer of machine data using the file is based on the channel principle:

HYDRA knows channels; each channel is configured in such a way as to assign an identifier from the file including its value to this channel. The channels are unique for each terminal. But the identifiers may also be assigned to several channels. Building groups in the configuration may define which channel belongs to e.g. which machine or unit. Each group requires different communication files!

HYDRA provides the below-mentioned channel types – please also refer to the document entitled “Prerequisites_HYDRA_Machine_Connection.pdf“ for further information:

| | |
|---|---|
| C: | Counter channels (ascending, persistent machine counters with specified overflow) |
| Z: | Cycle channels to collect cycle times of the machine |
| I: | Digital input channels |
| O: | Digital output channels |
| MSTAT: | Numeric machine status, as an alternative to the digital input channels |
| P: | Process data channels |
| T: | Trigger channels triggering process data collection |

V: Value channels for specific read and write operations, e.g. scale requests or PCC-ADP information

These channels are numbered, pooled to groups by an INI file and named accordingly. The PCC-DIF module may be used in addition to other PCC protocol modules for each terminal. The modules in use are defined in the pccdll.ini file. All of these modules communicate with each other using this channel interface. Therefore, channels must be unique for each terminal. The respective configuration file of the modules specifies which channel belongs to which module.

# 3 Configuration of the File Interface

## 3.1 Conditions / Quantities

The details mentioned below have to be taken into account when PCC-DIF is configured or used:

| | | **Number of characteristics/process parameters for each machine** | **Number of machines** | **Fastest recording cycle** |
|---|---|---|---|---|
| **PCC-DIF** | Automatic | Up to 300 pieces | Up to 20 pieces | - |
| | Cyclic | Up to 300 pieces | Up to 20 pieces | 5 seconds |
| | | When it comes to cyclic recording, the quantity of characteristics is restricted in proportion to the input rate. | | |

## 3.2 PCC Configuration (pccdll.ini)

The module is registered in the PCCDLL.INI file:

driver=OPCMPDV.DLL

Example:

```
[SERVICE]
tracing=1
ShowErrorWindow=0

[DRIVER_1]
driver=OPCMPDV.DLL

[DRIVER_2]
driver=PCCDIF.DLL
```

## 3.3 Module Configuration (pccdif.ini)

The configuration is divided into the sections [SERVICE] as well as in up to several channel groups [GRP_001, …]. The service section includes general configurations for the module. The single groups include the access paths and channels of these groups.

A group includes:

INFILE Name of the input file with full path

OUTFILE Name of the output file with full path (the output is completely deactivated if nothing is indicated).

OUTFILE_DELETETIME Wait time in minutes after which the OUTFILE file is deleted. This prevents the file from increasing exceedingly when it comes to interferences in the communication. It goes without saying that in this case, the postings included in the file get lost.

INFILE_TIMEOUT Wait time in seconds after which a new INFILE file has to be provided by the communication partner. If this is not the case, an error message will be sent to the HYDRA terminal. If required, the terminal may trigger a response (posting, or similar). But the communication partner may also just send an empty file to indicate that it is still “alive”.

OUTFILE_TIMEOUT Wait time in seconds after which the OUTFILE file has to be picked up by the communication partner. If this is not the case, an error message will be sent to the HYDRA terminal. If required, the terminal may trigger a response (posting or similar).

OUTFILE_MINIMUMINTERVAL Interval time (seconds) after which a file has to be sent at the latest. HYDRA sends an empty file if there is no communication.

TMPINFILE Temporary input file

TMPOUTFILE Temporary output file

POLLTIME Interval for file polling in milliseconds.

PDV_MODE=AUTO Identified HYDRA-PDV values are immediately forwarded to data collection. The “MAN” (manual) mode enables cyclic or triggered transfers. Please also see section 2.4

SAVE_COUNTER=ON Current counter readings are saved when the application is finished. When the application is restarted, the user decides through selection at the terminal, whether or not the saved values are taken into account for the calculation of quantities.

SAVE_INTERVAL=60 Period of time for the cyclic backup of current meter readings (in seconds), provided that the application is not finished properly.

OVERFLOW=30000 The counter register increases up to a meter reading OVERFLOW specified in the INI file of the driver and then restarts from an initial value.

OVERFLOW_MODE=0 „0“ after an overflow, it is restarted at “0”.

„1“ after an overflow, it is restarted at “1”.

„RESET“ If a lower figure is suddenly recognized the last imported figure is considered being the overflow and the meter readings until reaching this overflow are ignored.

The channels

C:Cxxx Counter channel with the ordinal number xxx. Up to 999 channels may be configured

Z:Zxxx Cycle time of the counter channel xxx

I:Ixxx / O:Oxxx Digital channels

P:Pxxxx Process data channel with ordinary number xxxx

T:Txxxx Trigger channel for process values of the current group. A counter value or any other data value may be used here. Changes are responded to.

V:Vxxxx Value channel for specific read and write processes

Further channel configurations are possible on request.

Please note: Channel numbering including all protocol modules configured at the terminal must be unique.

Sample configuration (corresponds to the above-mentioned sample row):

[SERVICE]

info=PCCDIF.DLL

LogLevel=2

```
STARTUPTIME=1000

[GRP_001]

INFILE= C:\CTWIN\ DATA\M100_MDE.DAT
TMPINFILEEXTENSION =TMP

OUTFILE=C:\CTWIN\ DATA\OUTFILE_M100.DAT
TMPOUTFILEEXTENSION =TMP

SENDALLRECEIVEDVALUES=OFF
SEND_RECORDINDEX=OFF

OUTFILE_DELETETIME =60
INFILE_TIMEOUT=120
OUTFILE_TIMEOUT=120
OUTFILE_MINIMUMINTERVAL =60

PDV_MODE=MAN
SAVE_COUNTER=ON
SAVE_INTERVAL=60

POLLTIME=10000

BAS.DAT=DATE
BAS.ZEI=TIME

OVERFLOW=30000
OVERFLOW_MODE=0

;// Parameter

C:C001=COUNTER1
Z:Z001=ZYK1
C:C002=COUNTER2
I:I001=INP01

P:P0001=P1
P:P0002=P2
T:T0001=COUNTER1

V:V0001=WERT01
V:V0002=AUNR
```

## 3.4 Notes on the IDs BAS.DAT and BAS.ZEI:

DAT and ZEI should be used as ID in input data..

The IDs provided by the data record have to be used in order for the input data to be sent to the application. .

BAS.DAT=DAT

BAS.ZEI=ZEI

Configuration of the format in the file as of version 7.2.1.7

DATE_FORMAT=mm/dd/yyyy